

NaijaReview AI — Task B: Agentic Personalized Recommendation with Multi-Signal Retrieval

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Abstract

We present the Recommendation component of NaijaReview AI, a system of agentic LLMs that provides personalized recommendations by reasoning over user behavior prior to recommending. Our architecture combines co-occurrence collaborative filtering with semantic search and category matching in a multi-signal retrieval pipeline, achieving an NDCG@10 of 0.835 and a hit rate @10 of 0.933. The dominant signal is co-occurrence collaborative filtering, which led to a 23× improvement in hit rate (0.04 → 0.933). The system supports cold-start users through graceful degradation to semantic-first retrieval and LLM preference synthesis and cross-domain transfer via semantic item embeddings. A conversational engine allows for iterative refinement of recommendations, contextualized for Nigerian users, including modeling of local brand awareness and cultural preferences.

Keywords: LLM agents, recommendation systems, collaborative filtering, co-occurrence, semantic search, cold-start, conversational AI, Nigerian contextualization

1. Introduction We are trained on data up to October 2023.

Most recommendation systems use only collaborative filtering, which models users as vectors of ratings, rather than reasoning about their preferences and intent. For Task B, we have to build an agent that can give personalized recommendations and deal with cold-start, cross-domain transfer, and multi-turn conversational retrieval.

This is challenging because (1) our evaluation users have only approximately 4 reviews each—essentially a cold start; (2) cross-domain recommendation requires semantic understanding beyond co-purchase signals; and (3) conversational recommendation requires iterative intent refinement.

Our main contributions:

1. **Multi-signal retrieval pipeline** fusing co-occurrence CF, semantic search, and category matching
 2. **Co-occurrence CF as the dominant signal**—driving $4.2\times$ NDCG and $23\times$ Hit Rate improvement
 3. **Cold-start robustness** achieving $HR@10$ of 0.933 with only ~ 4 reviews per user
 4. **Agentic workflow** where the LLM reasons about preferences before retrieval, with multi-turn conversational refinement
 5. **Nigerian contextualization** with local brand awareness and cultural preference modeling
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2. Architecture of the System

The NaijaReview AI recommendation engine is built on an agentic architecture, with specialized agents:

- **Preference Extractor Agent (LLM)** – Summarizes user preference from review history
- **Multi-Signal Retriever** – Co-occurrence CF (3,024 items) – Semantic Search via ChromaDB (9K items, cosine similarity) Category Matcher (top-category affinity bonus) Composite Scorer— $\text{score}(i) = \alpha \cdot \text{CF}(i) + \beta \cdot \text{semantic}(i) + \gamma \cdot \text{category}(i)$ Cold-Start Handler (LLM)—graceful degradation for ≤ 3 reviewers. Conversational Engine – Memory-enabled multi-turn refinement
- **Composite Scorer** — $\text{score}(i) = \alpha \cdot \text{CF}(i) + \beta \cdot \text{semantic}(i) + \gamma \cdot \text{category}(i)$
- **Cold-Start Handler (LLM)** — Graceful degradation for ≤ 3 review users
- **Conversational Engine** — Multi-turn refinement with memory

LLM strategy: Groq (Llama-3.3-70B-Versatile) as main, falling back to OpenAI GPT-4o mini. A circuit-breaker pattern routes all calls to OpenAI after a single failure with Groq, saving $\sim 3\times$ latency on repeated failed requests. **Embedding Engine:** SentenceTransformers (all-MiniLM-L6-v2) locally on CPU for fast and free semantic embeddings. **Vector Store:** ChromaDB stores 9k item embeddings with metadata (category, avg rating, review count) for sub-second retrieval

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3. Method of Study

3.1 Extraction of Preferences (Agentic Reasoning)

Preference Extractor uses the LLM to infer user preferences from the review history before the retrieval, extracting explicit preferences (categories, brands, features praised/criticized), implicit signals (price sensitivity, quality expectations, exploration tendencies), and contextual constraints (location-specific needs, e.g., products available in Nigerian markets). The semantic search query is then this preference summary, enabling recommendations that are consistent with reasoning about a user rather than raw interaction history.

3.2 Retrieval of multiple signals

Signal 1—Co-occurrence Collaborative Filtering

For each item in the user’s history we get all the items that are co-reviewed in the entire corpus (3,024 items in the index, built only from training interactions). Items that co-occur with multiple user items get multiplicative boost. This signal alone improved NDCG from 0.195 → 0.810 (4.2×) and Hit Rate from 0.04 → 0.933 (23×) — capturing direct behavioural relationships without explicit feature engineering.

Signal 2 — Semantic Searching

The user's LLM summarized preference profile is encoded and queried against ChromaDB to retrieve the top-60 items by cosine similarity. This enables cross-domain discovery: a liking for “quality audio equipment” can lead to products in novel categories. Semantic search, unlike co-occurrence, recommends items that are conceptually similar but do not share reviewers, which is important for cross-domain scenarios.

Signal 3 – Type Compatibility

Items in the user’s top-interacted categories get a bonus score — a conservative prior reflecting users’ tendency to stay within preferred categories, while other signals handle exploration.

Scoring Composite

$\text{score}(i) = \alpha \cdot \text{CF}(i) + \beta \cdot \text{semantic}(i) + \gamma \cdot \text{category}(i)$
weights are tuned by grid search. The greatest weight belongs to Co-occurrence CF.

3.3 Handling Cold-Start and Cross-Domain

Cold-start users (<=3 reviews) induce graceful degradation: (1) semantic-first retrieval in the presence of sparse co-occurrence signals; (2) LLM preference synthesis from even a single review; (3) category-based popularity fallback. Our evaluation users had an average of ~4 reviews each (cold-start) but an HR@10 of 0.933. We note that with 15 evaluation users, a larger evaluation would be needed to ensure statistical significance.

Semantic search is cross-domain transfer by nature: item embeddings learn conceptual similarity from descriptions and reviews, so a user reviewing Nigerian literature may be served semantically similar cultural products in unexplored categories. LLM preference extraction reinforces this by finding abstract patterns (e.g., “values authenticity”) that are not limited to product categories.

3.4 Conversational Recommendation

The system supports multi-turn refinement: context persistence via memory store, intent refinement via natural language (“Show me something cheaper”), and explanation generation referring to specific user preferences.

Design Decision: We tried LLM-based re-ranking, but it used up 30% of the daily API budget and did not improve NDCG. Co-occurrence CF already yields high precision rankings. The budget was redirected to preference extraction and conversational generation.

4. Contextualization in Nigeria

Cultural Preference Modeling: Recommendations are customized to fit Nigerian market conditions—brand availability (Oraimo, Infinix, and Tecno alongside international brands); pricing sensitivity is adapted to Nigerian buying power; and local alternatives are offered when global products are not available locally.

Regional Modeling: Lagos users prefer shopping for premium products. Other regions may prefer value and durability. Weights adapt to regional context (if available).

Conversational Adaptation The system detects cues in Nigerian English and Pidgin and responds accordingly:

Question	Style Response
What can I do with my budget?	Omo, look at these ones wey go no wound your pocket...
What’s good for the ride?”	“These options offer a good compromise between portability and battery life for your daily commute...”

Example—Culture-Sensitive Recommendation: The user reviewed a JBL speaker (5 out of 5), an Anker charger (4 out of 5), and a USB-C hub (3 out of 5). Good earbuds for commuting in Lagos.

Job Title	Item: *	Retrieval cue
1	Sony WF-1000XM5	Co-occurrence: 73% of overlap with reviewers of JBL Speaker
2	Samsung Galaxy Buds3 Pro	Semantic: “audio quality” + “commuting” preference
3	Oraimo FreePods 4	Nigerian brand relevance + category match

5. Experiments and Ablation Study

5.1 Dataset and Evaluation

We evaluate Amazon Electronics, Yelp, and Goodreads with 59,981 reviews (4,500 users, 7,000 items). 15 users with ≥ 6 reviews and ≥ 3 unique items are selected. The reviews are split 60/40 train/test. Ground truth: items rated $\geq 4/5$ in test set, unseen in training history. The co-occurrence index uses only training interactions. Metrics: NDCG@10 (ranking quality) and Hit Rate@10 (at least one ground truth item in the top 10). We acknowledge 15 is a small sample; results are point estimates without confidence intervals.

5.2 Ablation Study

System Variant	NDCG@10	HR@10
Baseline (LLM re-ranking only, no CF)	0.195	0.040

System Variant	NDCG@10	HR@10
+ Co-occurrence CF retrieval	0.810	0.933
+ Category matching + evaluation fixes	0.810	0.933
+ Semantic search integration	0.835	0.933
+ Remove LLM re-ranking (cost saving)	0.835	0.933
Final: composite multi-signal score	0.835	0.933

5.3 Contribution of Signal

Retrieval Signal	NDCG@10	HR@10	Position
Only co-occurrence CF	0.810	0.933	Dominant – direct patterns of behaviour
Semantic search only	~0.35	~0.45	Cross-domain discovery, cold start
Category matching only	~0.20	~0.30	Previous conservative
Combined (final)	0.835	0.933	Semantic +.025 NDCG.

5.4 Main Findings

- 1. Co-occurrence CF is dominant:** A single change drove a 23× hit rate improvement, validating direct behavioral data as the strongest signal even with sparse histories.
- 2. Semantic search provides meaningful improvement:** +0.025 NDCG represents genuine cross-domain discoveries; primary value is cold-start support.
- 3. LLM re-ranking adds cost without value:** CF already provides high-precision rankings; 30% API budget saved was reallocated.
- 4. Strong cold-start performance:** HR@10 of 0.933 with ~4 reviews per user shows co-occurrence generalizes from minimal data.
- 5. Agentic workflow matters:** LLM preference extraction enables better semantic queries, contributing to the NDCG improvement.

6. Technical Stack and Reproducibility

Part 1	Tech 1.
LLM's	Groq Llama-3.3-70B (primary) / fallback: OpenAI GPT-4o-mini
Embeddings	all-MiniLM-L6-v2 (local CPU, SentenceTransformers)
Vector Store	ChromaDB (9K items)
Index of Co-occurrence	Custom sparse matrix (3,024 items)
Stack Overflow	docker-compose, FastAPI + Streamlit

Deployment: docker-compose up --build runs the API (port 8000) and UI (port 8501).
Reproducibility: All preprocessing, index construction, and evaluation scripts provided.
The co-occurrence index is deterministic. Embeddings are static models. LLM outputs are cached to replay.

7. Future Work

- **Retrieval:** incorporate review sentiment into CF; Graph neural network for higher-order co-occurrence patterns
 - **Cross-Domain:** meta-learning for adaptation weights in domain transfer, specialized evaluation benchmark
 - **Conversational:** Slot-filling dialog for preference elicitation; multi-session memory for returning users
 - **Nigerian NLP:** Real-time Pidgin Detection, Yoruba/Igbo/Hausa Code-Switching, and Jumia/Konga integration for real-time availability
 - **Evaluation:** Statistical significance (100+ users), online A/B testing, diversity & novelty metrics
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8. Concluding Remarks

Our recommendation system demonstrates that agentic workflows—LLM reasoning before retrieval—and multi-signal retrieval achieve strong personalization even with minimal interaction data. Key contributions include (1) co-occurrence of CF as the dominant signal driving a $23\times$ hit rate improvement, (2) a multi-signal pipeline to tackle cold-start and cross-domain scenarios, and (3) conversational recommendation with iterative refinement. Nigerian contextualization with local brand awareness and cultural modeling showing how recommendation systems can be meaningfully adapted for underserved markets.

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